

$\Rightarrow$ Type I & Type II Errors

$\hookrightarrow$ Pre-req: Confusion Matrix

	1	0
1	TP	FN
0	FP	TN

Type II

Type I

Type I:

- $\rightarrow$ One person is tested HIV -ve yet the system treats the person HIV +ve.
- $\rightarrow$ System flags as not spam but turns out to be spam.

Type II:

Type II :

- Person is HIV+ve & system says HIV ve.
- System flags as spam but is not spam.

=> Application: TPR & FPR

→ Sensitivity : True Positive Rate

$$TPR = \frac{TP}{TP + FN}$$

→ Specificity : $\frac{TN}{TN + FP}$

→ False Positive Rate: FPR

$$FPR = 1 - \text{Specificity}$$

$$FPR = 1 - \frac{TN}{TN + FP}$$

or

$$FPR = FP$$

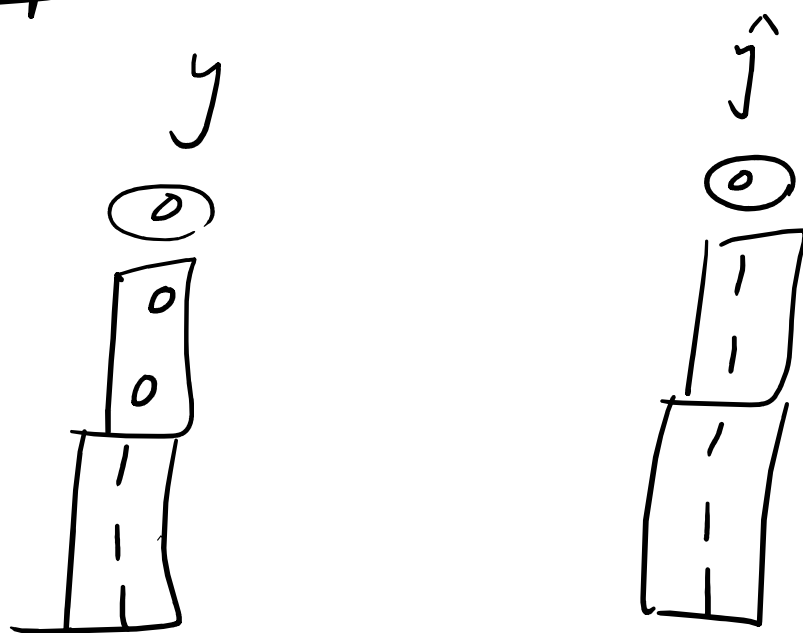
or

$$FRL = \frac{FP}{TN + FP}$$

=> Specificity when high Type I Error lowest.

=> Sensitivity when high Type 2 Error lowest.

=> Example:



TP	FN
FP	TN

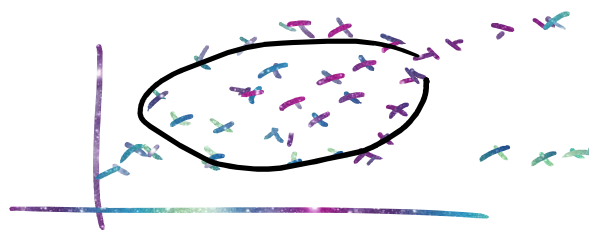
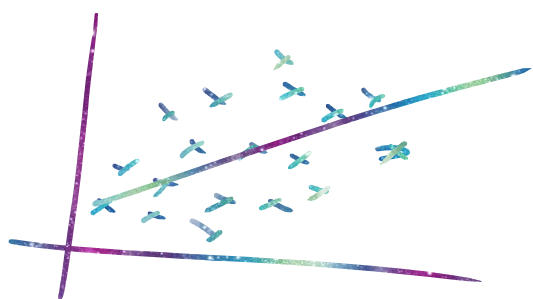
3	0
2	1

$$\Rightarrow \text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{1}{1 + 2} = \underline{\underline{33.33\%}}$$

$$\Rightarrow \text{Sensitivity} = \frac{TP}{TP + FN} = \frac{3}{3 + 0} = 100\%$$

$\Rightarrow$ Parametric $\int$ Non-parametric
 (Pre-calculating) (there is no calculation of parameters)



Naive Bayes : Pre-requisites

$\Rightarrow$ Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$\Rightarrow$ Two dices are rolled:

$\{(1,1)(1,2)(1,3)(1,4) \dots \dots \dots$
 $(2,1)(2,2) \dots \dots$
 $(6,6)\} \quad n(s) = \underline{\underline{36}}$

		D1					
		1	2	3	4	5	6
D2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

$\Rightarrow$ What is the probability that $D_1 = 5$
 $\& \ D_1 + D_2 \leq 10$

$$= \frac{5}{36}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$= \frac{5}{36} \div \frac{1}{6}$
 $= \frac{5}{36} \times \frac{6}{1}$
 $= \frac{5}{6}$

$\Rightarrow$ Independent Events :

⇒ Independent Events :

$$P(A \cap B) = P(A) \times P(B)$$

⇒ Mutually Exclusive Events :

A, B

$$P(A \cap B) = 0$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A|B) = 0$$

⇒ Bayes Theorem :

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

↓
Likelihood

Prior Prob

↑
Evidence

→
Posterior

⇒ Example :

Consider a manufacturing units of dipsets -

Consider a manufacturing mix of chips -

$$\begin{array}{c} M_1 \\ (20\%) \\ \hline 5\% D \end{array}$$

$$\begin{array}{c} M_2 \\ (30\%) \\ \hline 3\% D \end{array}$$

$$\begin{array}{c} M_3 \\ (50\%) \\ \hline 1\% D \end{array}$$

Question: What is the prob of being defective
from M_3 .

$$\Rightarrow P(M_1) = \frac{1}{5} \quad P(M_2) = \frac{3}{10} \quad P(M_3) = \frac{1}{2}$$

$$P(M_3/D) = \frac{P(D|M_3) \times P(M_3)}{P(D)}$$

$$P(D|M_3) = \frac{1}{100} \quad P(D|M_2) = \frac{3}{100} \quad P(D|M_1) = \frac{5}{100} \\ = \frac{1}{20}$$

$$P(D) = P(D \cap M_1) + P(D \cap M_2) + P(D \cap M_3)$$

$$P(D) = P(D|M_1) \times P(M_1) + P(D|M_2) \times P(M_2) \\ + P(D|M_3) \times P(M_3)$$

$$= \frac{1}{20} \times \frac{1}{5} + \frac{3}{100} \times \frac{3}{10} + \frac{1}{100} \times \frac{1}{2}$$

$$= \underline{\underline{0.024}}$$

$$\therefore P(M_3 | D) = \frac{P(D | M_3) \times P(M_3)}{P(D)}$$

$$= \frac{\frac{1}{100} \times \frac{1}{2}}{0.024} = \underline{\underline{20.83\%}}$$